Midterm 1, Math 170A - Section 1, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: ____

Signed name: ____

Student ID number: ____________________________________

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- You final answers does not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. This problem only: please provide answers with very short explanations.

   (a) (2 points) For any event A what is $\mathbf{P}(A) + \mathbf{P}(A^c)$?

   (b) (2 points) Let A and B be events with strictly positive probabilities. Assume that their probabilities are not equal. If $\mathbf{P}(A|B) = \mathbf{P}(B|A)$, what is the value of $\mathbf{P}(A \cap B)$?

   (c) (2 points) Suppose that $B^c \subset A$, A and B are independent. What can you say about $\mathbf{P}(A)$ and $\mathbf{P}(B)$?

   (d) (2 points) Assume that events A and B are independent. Express $\mathbf{P}(A \cup B)$ in terms of $\mathbf{P}(A)$ and $\mathbf{P}(B)$.

   (e) (2 points) Give an example of three sets A, B and C, so that A and B are independent, but A and B become dependent, given C. Make sure to specify the probability space (outcomes and probability law).

2. (10 points) Electric light bulbs are made at three plants:
Plant 1 (40%), Plant 2 (?? %) and Plant 3 (?? %).
The defective rates are:
Plant 1 (10%), Plant 2 (20%) and Plant 3 (5%).
Overall the defective rate is 12%. Find the proportion of light bulbs made at Plant 2.

3. (10 points) There are 57 lockers in a dark hall. On a rainy day 30 students come independently and put umbrellas in these lockers randomly without looking inside (every student has one umbrella and chooses a locker uniformly at random). What is the probability that there is a locker with more than one umbrella?

4. (10 points) Let $n \geq 4$ be a positive integer. We are given a deck of n cards labeled with numbers $1, 2, \ldots, n$, and we select uniformly at random two cards from this deck. Each of these two selected cards is destroyed with probability $1/2$, independently of the other card. What is the probability that neither card number 1 nor card number 2 get destroyed?

5. (10 points) There are two stores in your neighborhood selling lottery tickets. The first store sells four types of lottery tickets priced at \$1, \$2, \$3 and \$4. The second store sells five types of lottery tickets priced at \$1, \$2, \$3, \$4 and \$5. The first store is closer to your house, so you go there two times as often (namely, the probability that you decide to go to the first store is two times probability you decide to go to the second store).

   One day you had \$8 and you spent them all on 3 lottery tickets, each of different type, chosen uniformly at random in some store. What is the probability that you were in the second store?